

Introduction

The modern office or home, as far as electrical/electronic appliances are concerned, differs primarily from one ten years ago by the fact that wires have almost disappeared for all practical purposes. Gone is the age of desks cluttered with tangled wires. Today, except for power transmission wires, all wires can be done away with. The internet is wireless, phones are wireless, computer peripherals are wireless. The last frontier, wireless power transfer, is soon about to be breached. The time seems ripe for us to delve into the world of wireless technologies, and the first logical question in this direction seems to be to ask the exact definition of wireless.

1.1 What is wireless?

Put simply – a wireless technology is any technology in which wires have been eliminated in areas where wires were required before. Almost all wireless applications have been in the field of communication (more specifically, in transferring signals and not just communication between humans). Examples range from radio and TV broadcasting to mobile phones, Bluetooth and wireless LAN.

While almost all developments in this direction have been made possible using electromagnetic waves (explained in detail later), some very specific applications have also been developed using ultrasonic (high frequency sound) and infrasonic (low frequency sound) waves.

Before we go into how exactly wireless communication takes place, let us take a brief look at its history.

1.2 History of wireless technologies.

Though the scientific world had been aware of both electricity and magnetism separately for a long time, the connection between the two had not been noticed until the 19th century. Reports in the early 19th century about the connection were largely unnoticed by the scientific community. In 1820, Hans Christian Oersted accidentally discovered that a current carrying wire caused a magnetic needle in a compass to deflect, and became the first to record evidence of the relationship